MILITARY SPECIFICATION

GAGE , DEPTH, HYDROGRAPHIC , NON-MAGNETIC

SELF - LUMINOUS DIAL

1. SCOPE

1.1 This specification covers a non-magnetic pressure gage with self-luminous dial for use in shallow sea water.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-M- 500- Molding Plastic, Methacrylate.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape; Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing).

MILITARY

MIL-T-704- Treatment and Painting of Material.
MIL-E-5272- Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-E-7989 - Covers, Light-Transmitting, for Aeronautical Lights, General Specification for.
MIL-C-15328 - Coating, Pretreatment (Formula No. 117 for Metals).
MIL-R-15624 - Rubber Sheets; and Cut, Molded and Extruded Special Shaped Sections - Synthetic, Medium Soft, Shipboard Gasket Use Except Low Temperature Application.
MIL-P-15929- Primer Coating, Shipboard, Vinyl-Red Lead (Formula No. 119 for Hot Spray).
MIL-E-15935 - Enamel, Outside, Gray, Vinyl-Alkyd (Formula No. 122-11).
MIL-M-19590- Marking of Commodities and Containers to Indicate Radioactive material,
MIL-M-19595 - Magnetic Effect Limits for Nonmagnetic Equipment Used in the Proximity of Magnetic Influence Ordnance.

STANDARDS

MILITARY

MIL-STD-8 - Dimensioning and Tolerancing.
MIL-STD-10- Surface Roughness, Waviness, and Lay.
MIL-STD-105- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129- Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS

810-1895181- Carrying Case & Gage Assembly
810-1895182- 1) Gage Assembly 2) Calibration Assembly
810-1895183- Miscellaneous Details

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DRAWINGS (Contd)

BUREAU OF SHIPS
810-1895184- Carrying Case Assembly & Details
810-1895185- Carrying Case Details
810-1895186- Lanyard Assembly&Details
810-1895187- Gage Base Assembly& Details
810-1895188- Diaphragms
810-1895189- Movement Assembly& Dial Glass Details
810-1895190- Pointer Assembly & Details
810-1895191 - Back Cover Assembly & Details
810-1895192- Dial Assembly & Details
810-1895193- Sector Assembly & Details
810-1895194- Pinion Assembly & Details
810-1895195- Movement Frame Assembly & Details

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York, N. Y. 10016.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should bc addressed to the Superintendent of Documents, Government Printing Off ice," Washington, D.-C, 20402.)

3. REQUIREMENTS

3.1 Qualification. - The gages furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has **been** listed on or approved for listing on the applicable qualified products list.

3.2 Description. - The hydrographic depth gage is a precision instrument that accurately indicates the depth of sca water (specific gravity 1. 025), from a free surface down to a depth of 50 feet. The gage consists essentially of a pressure-sensing element, a movement mechanism and an indicating dial that are enclosed in an evacuated, pressureproof casing. The gage shall have a low magnetic effect; and, the function of the gage shall not be impaired by vibration, impact, underwater blast, hydrostatic pressure, or ambient temperature variations, as hereinafter specified. The gage shall be constructed in accordance with 810-1895181 to 810-1895195 inclusive, and as specified herein.

3.3 Material. - Material shall conform to specifications as indicated on applicable drawings Material not definitely specified shall be of good quality, and free from defects that will affect the intended use. Materials with minor variations of composition to specified materials will be acceptable, provided the mechanical properties of the material are not affected. Material shall meet the magnetic effect limits as specified in MIL-M-19595 (see 3.9 and 6. 6).

3.4 Indicating dial. - The indicating dial shall be the indicating hand type. The dial markings and graduations shall be silhouetted on a band of self-luminous radioactive compound in a sealed source. The applied compound shall emit a green hue, or other hue, as specified (see 6. 2); and shall not crack, peel, or chip. The sealed source container shall be uncolored, optically clear, acrylic molding plastic conforming to class 2, type I of L-M-500,

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3. 4.1 Dial graduation. - The dial shall be graduated in l-foot increments up to and including the 50-foot' depth, The graduations shall be marked with consecutive Arabic numerals.

3. 4.2 Radiological contamination. - Each indicating dial shall comply with the requirements for sealed sources as specified in 4.5, 1.

3. 4.3 Brightness. - The visible portions of the indicating dial shall be uniformly bright. Prior to final assembly with the gage, each dial shall undergo the brightness test specified in 4.5.11. The initial brightness shall be 17. 5±2. 5 microlamberts, Dials that fail to meet the initial brightness requirements shall not be used for the construction of gages that are presented to the Government for quality conformance inspection (see 4. 4). Subsequent to final assembly, the brightness shall be determined to be a minimum of 10 microlamberts. Assembled gages that fail to meet the brightness requirements will not be presented to the Government for acceptance,

3.5 Casing, - The case is composed of the dial window, the bezel guard, the gage base and the back cover.

3. 5.1 Dial window. - The dial window shall permit unobstructed view of the dial graduations and markings; and shall be clear, uncolored glass conforming to class C of MTL-C-7989. The window shall withstand the impact shock specified in 4.5, 7.

3. 5.2 Bezel guard. - the juncture of dial window and casing shall be protected with a synthetic rubber guard securely cemented to the casing (see 4.5. 4). Synthetic rubber shall conform to class 1 of MIL-R-15624. The guard shall match the color specified in 3. 16.3 (see 3.16. 4),

3. 5.3 Mounting eyes. - The casing shall be equipped with integral mounting eyes suitable for mounting the gage on a flat surface or for carrying the gage on a lanyard. The eyes shall be a folding or retractable type such that when not in use they may be locked flush with the gage.

3.6 Lanyard, - A lanyard conforming to 810-1895186 for carrying the gage suspended from the divers neck shall be furnished with each gage, The color of the lanyard shall match the color specified in 3. 16.3 (see 3. 16.4,)

3.7 Carrying case. - Each gage shall be equipped with a carrying case conforming to 810-1895184 and 810-1895185. The exterior of the case shall match the color specified in 3, 16.3 (see 3.16. 4).

3.8 Radiological contamination. - Radiological contamination of the gage shall be within the limit as specified in 4.5.1.

3. 8.1 The manufacturer of the radioactive sources shall comply with the applicable A. E. C. regulations governing sealed sources containing beta or gamma-emitting radioisotopes.

3.9 Magnetic effect, - The gage shall comply with the magnetic effect limits specified in MIL-M-19595 (see 4.5. 2).

3, 10 Vibration. - The gage shall be capable of withstanding the vibration shocks specified in 4.5.5.

3.11 Impact shock. - The gage shall be capable of withstanding the impact shock specified in 4.5.6.

3.12 Underwater blast shock. - The gage shall be capable of withstanding the underwater blast shock specified in 4.5, 8.

3.13 Hydrostatic pressure. - The gage shall be capable of withstanding the hydrostatic pressure specified in 4.5.9.

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3.14 Accuracy. - At any instant, at any dial position, and at any depth, within the range specified herein; and at any ambient temperature between 32° and 90°F; the gage shall indicate the depth of submergence in sea water (specific gravity 1. 025) with an accuracy of plus or minus O. 50 foot in the zero to 50 foot range, inclusive (see 4.5. 12).

3.15 Identification marking. - The casing of each gage shall be legibly and permanently stamped, engraved, or etched with the following identification:

USN
Hydrographic Depth Gage, X-1
Serial No.

3. 15.1 Serial number. - The serial numbers shall be consecutive Arabic numerals beginning with 1 and ending with the number of gages on order, followed by the year of manufacture, and manufacturer's name or trade-mark, thus:

Serial No. 123/1965/Doe Co.

3. 15.2 Radiation symbol and non-magnetic marking. - The radiation symbol and non- magnetic marking shall be legibly and permanently on the indicating dial in a manner that does not obscure or conflict with the readability of the dial graduations. The radiation symbol shall conform to MTL-M-19590. The non-magnetic marking shall be:

"Non-Magnetic

3.16 Treatment and painting. - Metallic surfaces of the gage that are exposed to sea water shall be treated and painted in accordance with MIL-T-704, except as specified herein.

3. 16.1 Cleaning. - The surfaces to be painted shall be cleaned in accordance with the procedure specified in MIL-T-704 for ferrous metal surfaces.

3. 16.2 Treatment.

3. 16.2.1 Pretreatment. - Immediately after cleaning, a pretreatment coat complying with MIL-C-15328 shall be applied to the surfaces to be painted in accordance with the procedure specified in MIL-T-704 for the application of paint. The pretreatment coat shall be allowed to air dry for at least 1 hour.

3. 16.2.2 Priming. - When the pretreatment coat is dry, one coat of primer conforming to MIL-P-15929 shall be applied to the surfaces to be painted in accordance with the procedure specified in MIL-T-7W for the application of paint. The primer coat shall be allowed to air dry for at least 1 hour.

3. 16.3 Finishing. - The surfaces treated in accordance with 3.16.2, 1 and 3, 16. 2.2 shall be finished with two coats of outside gray paint conforming to MIL-E-15935, applied in accordance with the procedure specified *in* MIL - T- 704 for the application of paint. The initial finish coat shall be allowed to air dry at least 1 hour before the application of the final coat. The final coat shall air dry for at least 168 hours, or shall be force dried at 1250 to 150" F. for at least 6 hours before presenting the gages for acceptance inspection.

3, 16.4 Uniformity of colors. - The painted surfaces of the gage, the bezel guard, the carrying case, and the lanyard of each gage shall be of a uniform color.

3.17 operating and maintenance instruction. - An instruction sheet or sheets, as is necessary, describing the detailed procedures for the operation and maintenance of the gage shall be furnished with each gage (see 5. 3). The instructions shall include diagrams, figures, or photographs that are necessary to illustrate the procedures. Prior to final delivery, 'the instructions shall be submitted to the Bureau of Ships for approval,

3, 18 Repair parts kit. - When specified (see 6. 2), a repair parts kit shall be furnihsed. This shall consist of:

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2 each lanyards
60 each brass mounting screws
60 each brass bolts with nuts
30 each bottom screws
30 each mounting eyes bolts
30 each mounting eyes
2 each rubber bezel guards

3.19 Screw threads, - Screw threads shall be American National Standard in accordance with Handbook H28.

3.20 Finished surfaces. - When finished surfaces are specified on the drawing, the surfaces shall conform to MIL-STD-10. (Dimensional and tolerance symbols are in accordance with MIL-STD-8.)

3.21 Workmanship. - Workmanship shall conform to the standards of accepted commercial practice. The exterior of the gage shall be smooth and free of projections that will injure human skin or puncture lightweight pure gum sheet rubber,

4. QUALITY ASSURANCE PROVISIONS

4, 1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification; where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests, - Qualification tests shall be conducted at a laboratory satisfactory to the Bureau of Ships. Qualification tests shall consist of the tests specified in 4.5. Application for Qualification tests shall be made in accordance with “Provisions Governing Qualification” (see 6.3 and 6. 4).

4.3 Sampling for quality conformance inspection. -

4.3.1 Lot. - For the purpose of quality conformance inspection, a lot shall consist of all the gages presented at one time that have passed the radiological contamination and magnetic effect test of 4. 5.1 and 4,5.2.

4.3, 2 Sampling for quality conformance injection (except radiological contamination and magnetic effect). - A random sample shall be selected from each lot in accordance with MIL-STD-105, inspection level III, Gages so selected are hereinafter designated as the “acceptance sample”. Unless otherwise specified herein, the acceptable quality level (AQL) shall be O. 40.

4. 3.2.1 Sampling for vibration, impact, dial glass impact and hydrostatic pressure. - A “test sample” shall be selected from each lot (see 4.3. 2) as listed in table I. The sample shall be drawn from the quality conformance inspection sample (see 4.3. 2). A gage so selected shall hereinafter be designated as a “test gage”. Test gages shall be legibly marked in a conspicuous place by stencil or printing as test gage, Each test gage shall undergo the tests specified in 4.5, 5 to 4.5.12, inclusive, in the sequence of specification paragraphs. Unless otherwise specified, acceptance or rejection lots shall be in accordance with table 1, Upon satisfactory completion of all tests, test gages of the test sample shall be refinished and delivered as part of the lot.

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Table I - Sampling for vibration, impact, dial glass impact and hydrostatic pressure.

Number gages in lot (see 4.3. 1)	Number gages in test sample	Acceptance number	Rejection number
2- 8	2	0	1
9- 15	2	0	1
16- 25	2	0	1
26- 40	2	0	1
41- 65	2	0	1
66- 110	3	0	1
111- 180	3	0	1
181- 300	5	0	1
301- 500	5		
501- 800	7		
801- 1,300	10		
1,301-3,200	15		

4. 3.2.2 Number of g~es for accuracy test. - The number of gages to be tested for accuracy at each graduation, as specified in 4.5.12, shall correspond to the number of gages in the test sample as listed in table I. Twenty percent of the gages in each acceptance sample (see 4.3. 2) less the test sample or 10 gages, whichever is less, shall undergo an accuracy test under increasing and decreasing pressure as specified in 4.5, 12 for the 5, 15, 25, and 50 graduations only. Gages that fail to meet either of the foregoing accuracy tests shall be declared defective. Acceptance or rejection of the acceptance sample (4.3.2) for accuracy shall be in accordance with table L

4.4 Quality conformance inspection. - Each gage of the acceptance sample (see 4.3. 2) shall undergo visual examination (see~3), dimensional examination (see 4, 5. 4) and the dial brightness test (see 4.5. 11). The number of gages from the acceptance sample that shall undergo the accuracy test shall be as specified in 4.3, 2.2.

4.5 Test procedure. -

4. 5.1 Radiological contamination. - Radiological contamination of each sealed source and of each gage shall be determined in accordance with one of the following tests:

4. 5.1.1 Scrub test. - The object to be tested shall be immersed in a solution which will not attack the material of which the container or source holder is made and which has been demonstrated to be effective in removing the radioisotope involved when the object is scrubbed with a brush under the surface of the liquid. All surfaces of the object shall be brushed thoroughly under these conditions. The total radioactivity shall be measured in the residue obtained by evaporation of the solution.

4. 5.1.2 Immersion test, - The object to be tested shall be immersed in a solution which will not attack the material of which the container or source holder is made and which, under the conditions of this test, has been demonstrated to be effective in removing the radioisotope involved. The solution shall be heated to 50 ± 5° centigrade and held at this temperature for 8 hours. The total radioactivity shall be measured in the residue obtained by evaporation of the solution.

4, 5. 1.3 Wipe test. - All exposed external surfaces of the object to be tested shall be wiped thoroughly with a piece of filter paper of high wet strength and absorption capacity,. moistened with a solution which will not attack the material of which the container or source holder is made and which, under the conditions of this test, has been demonstrated to be effective in removing the radioisotope involved. The paper shall be allowed to dry. Then the radioactivity on the paper shall be measured.

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4. 5.1.4 Cellulose tape test, - This test is applicable to a device containing a sealed source in a source holder which has an opening through which the radiation emerges. This opening shall be kept covered for a minimum of 7 days with a piece of thin adhesive cellulose tape. The tape shall be removed carefully and the radioactivity which may have been deposited on its adhesive side shall be measured.

4. 5.2 Magnetic effect. - The magnetic effect limits of each gage shall be determined in accordance with MIL-M- 19595 (see 3.9 and 6. 6), Gages failing to comply with MIL-M-19595 shall be offered for delivery.

4. 5.3 Visual examination, - Each gage of the "test sample" shall be examined for obvious defects that will impair serviceability, such as uneven surfaces and rough edges; and for compliance with 3.15, 3.16, 3.20, 3.21, and 3.22. Gages with one or more defects shall be declared defective. The AQL for this examination shall be 1.0,

4. 5.4 Dimensional examination. - Each gage of the "test sample" shall be examined to determine compliance with 3.5. A force of approximately 1 pound shall be applied with the fingers; outwardly around the upper periphery, and upwardly around the bottom periphery of the guard. The guard shall not show evidence of separation from the casing. Gages with loose bezel guards shall be declared defective. The AQL for this examination shall be 1.0.

4. 5.5 Vibration shock. - Each "test gage" shall be subjected to the following vibration test. The gage shall vibrate for period of 5 minutes along each of the three mutually perpendicular axes at an acceleration of 10g and a frequency of 25 to 50 cycles per second (c. p,s.). The frequency shall be varied from 25 c. p. s. to 50 c. p,s. back to 75 c. p.s. in the period of 5 minutes, or at the rate of 10 c. p.s. per minute. Gages thal show evidence either of damage or of displacement of the indicator hand to the extent that the accuracy requirements of 3.14 cannot be met shall be declared defective.

4. 5.6 Impact shock. - Each "test gage" shall be subjected to the impact shock test in accordance with that part of procedure II covering equipment operation of MIL-E-5272 except: (a) the impact shock shall be 50g and (b) the impulse time shall be 20 ± 5 milliseconds. Three shocks shall be applied in each of the following positions:

(a) Gage horizontal, dial up.
(b) Gage on edge, dial vertical, 25 ft. graduation on the vertical axis.
(c) Gage on edge, dial vertical, 25 ft. graduation on the horizontal axis.

The impact shock shall not damage the gage. Gages that show evidence either of damage or of displacement of the indicator hand to the extent that accuracy requirements of 3.14 cannot be met shall be declared defective.

4. 5.7 Dial glass impact. - Each test gage of the "test sample" shall be placed flat on a horizontal bed of l-inch plywood, dial up with *the* bezel guard removed. Recesses shall be provided for the bracket of the gage to ensure even bearing of the case bottom with the horizontal surface. A 5/8-inch-diameter steel ball weighing approximately 0.565 ounce shall be freely dropped from a height of 39 inches: once, to strike the juncture of the dial glass and the bezel; and again, to strike the center of the dial glass. This test may be waived, at the option of the Government inspector, provided the contractor has furnished a warranty that the test has been successfully completed on a number of gages, at least equal to the number of test gages at the time of assembly of the dial glass with the gage casing, The dial glass shall not chip nor fracture under the shock of either impact. Gages with chipped or fractured dial glasses shall be declared defective. The bezel guard shall be recemented to the casing upon successful completion of this test.

4. 5.8 Underwater blast shock. - Each test gage of the test sample shall be subjected to the following underwater blast shock: In a relatively large, calm, deep body of open water, preferably the open sea, submerge the gage to a depth of at least 10 feet by suspension from a f lost or other means suitable for the purpose intended, The depth of water under the gage shall be not less than the depth of the water over the gage. Inshore from the gage, at a distance of 225 feet from the gage, submerge a charge of 4,4 pounds of HBX- 1, or explosive of equivalent impulse, so that the center of the charge is at the same depth as the center of the gage. Detonate the charge electrically. Subjected gages shall then meet the accuracy test of 4,5,12.

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4, 5.9 Hydrostatic pressure. - Each test gage shall be subjected to a hydrostatic pressure of 133.50± 1.35 pounds per square inch gage (psig) for a period of not less than 1 hour. The hydrostatic pressure shall not distort or damage the gage, cause the gage to leak or displace the index hand to the extent that the accu-. racy requirements of 3.14 cannot be met. Gages that show distortion, damage, leaks, or displacement of the index hand to the extent that the accuracy requirements of 3.14 cannot be met shall be declard defective.

4.5.10 Radiological contamination. - Upon successful completion of the tests of 4. 5.5 to 4. 5.9 inclusive, test gages shall undergo the radiological contamination test of 4.5.1. Gages that fail to meet the requirements of 4. 5.1 shall be declared defective. Lots represented by defective test gages shall not be offered for delivery.

4.5.11 Dial brightness. - Each test gage that has successfully passed the test specified in 4.5.10 shall undergo the following brightness test: Brightness of the assembled gage shall be determined photoelectrically. The photoelectric device shall have been calibrated not more than 6 months before the tests by a brightness standard that is certified for brightness by a Government Laboratory. Subject gages shall have been kept in complete darkness for at least 8 hours prior to and during the test. Gages that fail to meet the requirements of 3. 4.3 shall be declared defective.

4.5.12 Accuracy, - Each gage to be tested for accuracy (see 4. 3.2.2) shall undergo the following test: The test may be conducted at any ambient temperature between 32° and 90°F. Gage readings shall be reduced to normal atmospheric pressure, 29.92 inches of mercury. The gages shall be positioned at random attitudes varying from a dial horizontal position to a dial vertical position and subjected to an increasing hydrostatic pressure. The pressure may be simulated by a combination of pneumatic and hydrostatic pressure. The true pressure of sea water (specific gravity 1. 025), determined with a mercury manometer *or* other device of comparable accuracy, shall be recorded for each graduation of the gage dial. The pressure shall then be decreased and the true pressure again determined for each graduation on the dial. This cycle shall be repeated for a total of five times. The arithmetical means of the true pressures of the five cycles (reduced to normal atmospheric pressure) shall be employed to determine compliance with 3.14. Gages that fail to comply with 3.14 shall be declared defective.

4.6 Examination of preparation for delivery. - Preservation, packaging, packing and marking shall be inspected to assure compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - The wrist depth gage shall be individually protected and packaged in a folding or set-up paperboard box. Nonmagnetic material shall be used in the construction of the paperboard box, Operating and maintenance instructions shall be packaged together with the wrist depth gage in the unit package.

5.2 Packing. - Packing shall be level A, B, or C, as specified (see 6. 2).

5.2.1 Levels A and B. - Wrist depth gages, packaged as specified in 5.1 shall be packed in a fiberboard box conforming to class weather-resistant of PPP-B-636. Metal stitches shall not be used in the construction of the fiberboard boxes. Boxes shall be closed and sealed with minimum 1 1/2 inch-wide pressure sensitive tape conforming to PPP-T-76. The gross weight shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level C. - Wrist depth gages, packaged as specified in 5.1 shall be packed in a manner which will insure safe. delivery and acceptance at destination. Shipping containers shall comply with the Uniform Freight Classification Rules and Regulations or other regulations as applicable to the mode of transportation. Metal stitches shall not be used in the construction of the fiberboard boxes.

5.3 Packaging and packing of repair parts kit (see 3. 18). -

5.3.1 The screws and bolts shall be sealed in an envelope and placed with the lanyards and bezel guards in a paperboard container. The packaged items shall then be packed in fiberboard boxes conforming to class weather-resistant of PPP-B-636.

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5.4 Marking, - In addition to any special marking required by the contract or order and herein (see 5.4. 1), interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5. 4.1 Radioactive Material Marking. - Each interior and shipping container, containing wrist depth gages, shall be marked in accordance with the requirements of MIL-M-19590. Marking ink, paint *or other* material containing ferrous oxide or substances that do not conform to MIL-M-19590 shall not be used for marking unit containers.

6. NOTES

6.1 Intended use. - Gages covered by this specification are intended for usc by underwater swimmers doing hydrographic surveys and other associated tasks.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification,

(b) That the supplier of the luminous compound should furnish a warranty that the initial brightness of the luminous compound will not decrease by more than 20 percent over a period of 300 days when using thallium 204, or 25 percent when using PM147. (See 3.4.)

(c) Information required with serial number lists (see 3.15. 1).

(d) That the supplier should furnish a warranty that the strap is dyed uniformly and colorfast to match the color specified in 3.16, 3.

(e) Whether or not repair parts kit is required (see 3.18 and 6, 8).

(f) Government laboratory where underwater blast tests will be conducted (see 4.5. 8).

(g) Level of packing required (see 5. 2).

6.3 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-24147 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington, D. C. 20360, and information pertaining to qualification of products may be obtained from that activity, Application for Qualification tests shall be made in accordance with “Provisions Governing Qualification” (see 6, 4).

6.4 Copies of “Provisions Governing Qualification” may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pa. 19120.

6.5 At the time a contract is awarded, the Bureau of Ships will select a Naval Facility to perform the underwater blast test.

6.6 Materials specified in this specification normally conform to the requirements of MIL-M-19595 and 3.9, Variance in compositions or contamination during manufacture may produce a finished product that is unacceptable under the requirements of MIL-M-19595.

6.7 Radioactive isotopes are fission products obtainable from and under control of the Isotope Division, Atomic Energy Commission, Oak Ridge, Tennessee.

6.8 Normal requirement for repair parts kits is estimated as one kit for each 10 gages on order.

Preparing activity:
Navy - SH
(Project 4220-N066Sh)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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